

# Lucien Brey

MACHINE LEARNING CHEMIST

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## Working experience

### PhD Student

UNIVERSITY OF TÜBINGEN AND IMPERIAL COLLEGE LONDON

Oct. 2026 - Now

- ELLIS PhD student with Prof. Philipp Hennig (Tübingen AI Center) and Prof. Antonio del Rio Chanona (Imperial College London)
- Uncertainty aware scaling-up of chemical process with probabilistic machine learning

### Machine Learning Scientist

ATINARY TECHNOLOGIES INC.

Nov. 2024 - Jul. 2026

- Part of the AI Research and Innovation team building methodologies and algorithms for Bayesian optimization
- Develop and implement methods handling multi-objective, high cardinality discrete spaces and transfer learning setting
- Maintenance and optimization of probabilistic models (GPs, NPs, TFMs) and generative models
- Supervision of a master thesis from EPFL on LLMs in BO for Chemistry
- Responsible of academic collaboration: Prof. Hernandez-Lobato (Cambridge), Prof. Schwaller (EPFL) and Prof. Frei (York)

### Machine Learning and Data Science Intern

ATINARY TECHNOLOGIES INC.

Aug. 2023 - Jan. 2024

- Developing a LLM-based tool to enable database and literature search from user request for more efficient transfer learning
- Benchmarking and optimization of existing algorithms (GPs)

### Teaching Assistant

EPFL

- TA in the Advanced General Chemistry course for first year bachelor students in EPFL Life Sciences program and UNIL Geosciences 2020-2023
- Private teacher in Chemistry for a first-year gymnasium student Spring 2022

## Education

### École Polytechnique Fédérale de Lausanne (EPFL)

STUDENT

2019-2024

- MSc in Molecular and Biological Chemistry Sep. 2022 – Sep. 2024  
*Courses:* Machine Learning for Physicists, Artificial Neural Networks / Reinforcement Learning, AI for Chemistry, Computational Methods in Molecular Quantum Dynamics, Understanding Advanced Molecular Simulation
- BSc in Chemistry and Chemical Engineering Sep. 2019-June 2022  
*Courses:* Calculus I, Linear Algebra I, Probability and Stastics, Statistical Thermodynamics, Mathematical Methods in Chemistry, Quantum chemistry, Introduction to transport phenomena

### Massachusetts Institute of Technology (MIT)

MASTER THESIS

Mar. 2024 - Sep. 2024

- *Differentiable inverse design of metal oxides and perovskites catalyst via fine-tuning of surface oxygens site-properties* supervised by Prof. Rafael Gómez-Bombarelli and Prof. Philippe Schwaller
- Implemented a differentiable geometry optimization pipeline for site-properties in perovskite and metal oxide catalysts
- Investigating structure-properties relationship using machine learning interatomic potentials (MLIPs) as surrogate model (PaiNN and CHGNet)

### Kungliga Tekniska Högskolan (KTH)

EXCHANGE YEAR

Aug. 2021 - June 2022

- Exchange year abroad, discovering the culture, language and educational system in Sweden

## Publications

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- F. A. Formica, E. Wu, **L. Brey**, D. Pacheco Gutiérrez, R. Barbano, H. Tribukait, J. M. Hernández-Lobato, P. Laveille, L. M. Roch  
*Bridging Innovation and Efficiency: The Promises and Challenges of Self-Driving Labs as Sustainable Drivers for Chemistry*,  
Chimia, Vol. 79, Issue 9, 600-605 2025
- S. Gallarati, P. van Gerwen, R. Laplaza, **L. Brey**, A. Makaveev and C. Corminboeuf  
*A genetic optimization strategy with generality in asymmetric organocatalysis as a primary target*, Chemical Science, Vol. 15,  
Issue 10, 3640-3660 2024

## Skills

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Bayesian Optimization | Probabilistic Machine Learning | Gaussian Processes | Neural Processes | Flow  
**Skills** Matching Models | Transfer Learning | Reinforcement Learning | Organic Chemistry | Inorganic Chemistry |  
Quantum Chemistry | Molecular Dynamics  
**Programming** Python | PyTorch | BoTorch | GPyTorch | Numpy | Scikit-learn | Git | Docker | uv  
**Languages** French Native | English C2 | Spanish C1 | Italian B1

## Projects

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### EPFL, Prof. Philippe Schwaller

PROJECT STUDENT

Feb. 2023 - Jul. 2023

- Trained transformers (OpenNMT) on reaction SMILES for by-product prediction, completing USPTO-MIT dataset with stoichiometry and reaction conditions
- Developed a Streamlit web app for by-products prediction for safety concerns based on reference private database from a industrial partner

### EPFL, Prof. Philippe Schwaller

PROJECT DURING AI FOR CHEMISTRY COURSE

Apr. 2023 - May. 2023

- Built image classification pipeline for detecting irregularities in water leaves from jet pumps, trained using curriculum learning
- Developed an Streamlit web app for importing images and getting classification. Later, our code was used in a physical chemistry lab to setup the experiment

### EPFL, Prof. Clémence Corminboeuf

PROJECT STUDENT

Sep. 2022 - Jan. 2023

- Building database based on experimental results for enantioselectivity of catalyst/substrate couples for Pictet-Spengler reaction
- Using kernel ridge regression as fitness function for predicting enantioselectivity in the genetic algorithm

### KTH, Prof. Mårten Ahlquist

RESEARCHER ASSISTANT

Jan 2022-Jun.2022

- Mechanism rationalization of ligand-controlled product selectivity in electrochemical carbon dioxide reduction with manganese bipyridine catalysts using QM Jaguar methods
- Structure optimization, transition state search and 3D modelling on Schrodinger Maestro software

## Associations

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### EPFL Chemistry Student Society (AdEC)

PRESIDENT

Sep. 2022-Aug. 2023

- Organizing events such as Weekend Ski, Fondue de Section, Welcome Day for new students, quiznights in a bar etc.
- Coordination of seminars with the Swiss Chemical Society (SCS), professors, companies and alumni

### KTH THS International

BUDDY

Jan. 2022-Mar.2022

- Responsible of a group of international students, organizing, preparing and staffing during events such as museum visits or trivia nights

### Coaching Chimie EPFL

TREASURER

Sep. 2020-Jun. 2021

- In charge of the association expenses, expense report, budget etc.
- Organization of events (game nights, picnics...) for the first year students to help them socialize and get introduced to the student life